

**9 D** tension in string 1 provides centripetal force for 3 masses so must be largest magnitude; eliminate **A** and **C**

centre of mass of 3 separate masses is at centre i.e. mass 2 so tension in string 1 is equivalent to a string providing centripetal force to a mass of  $3m$  at radius  $2r$ .

$$T_1 = (3m)(2r)\omega^2$$

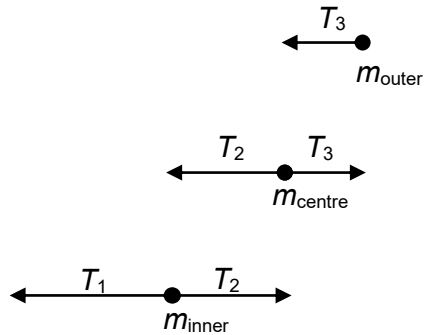
$$T_3 = m(3r)\omega^2$$

$$\frac{T_1}{T_3} = 2$$

eliminate **B**

**or**

via considering free bodies:



$T_3$  provides centripetal force on outer mass

$$m(3r)\omega^2 = T_3$$

vector sum of  $T_2$  and  $T_3$  provides centripetal force on centre mass

$$m(2r)\omega^2 = T_2 - T_3$$

$$\begin{aligned} T_2 &= m(2r)\omega^2 + T_3 \\ &= m(2r)\omega^2 + m(3r)\omega^2 \\ &= m(5r)\omega^2 \end{aligned}$$

vector sum of  $T_2$  and  $T_1$  provides centripetal force on inner mass

$$mr\omega^2 = T_1 - T_2$$

$$\begin{aligned} T_1 &= mr\omega^2 + T_2 \\ &= mr\omega^2 + m(5r)\omega^2 \\ &= m(6r)\omega^2 \end{aligned}$$